



Features

- SMD package with industry standard pinout
- Operating temperature range -40 ~ +90°C
- Comply to BS EN/EN55032 radiated Class B without additional components
- High efficiency up to 83%
- Protection: Short circuit
- 3KVDC I/O isolation
- Low cost
- Optional reel packing
- 3 years warranty

Applications

- Telecom/datacom system
- Wireless network
- Industrial control facility
- Instrument
- Analyzer
- Detector
- Data switch

GTIN CODE

MW Search: <https://www.meanwell.com/serviceGTIN.aspx>

Description

DETNO1 series is 1W isolated and unregulated module type DC-DC converter with SMD package. It features international standard pins, a high efficiency up to 83%, wide working temperature range -40~+90°C, 3KVDC I/P-O/P isolation voltage, compliance to BS EN/EN55032 radiated Class B without additional components, short circuit protection, etc. The models account for different input voltage 5V/12V/24V±10%, and various output voltage, ±5V/±12V/±15V for Dual output which are suitable for all kinds of systems, Such as industrial control, telecommunication field, distributed power architecture, and so on.

Model Encoding

DETNO1L-12SC

Performance difference

Output voltage (±5/±12/±15Vdc)

Input voltage (L: 4.5 ~ 5.5Vdc, M: 10.8 ~ 13.2Vdc, N: 21.6 ~ 26.4Vdc)

Rated wattage

Series name

Type	Description	Note
Blank	-40~+90°C working temperature with max. 1 second short protection	In Stock
SC	-40~+105°C working temperature with continuous short protection	Optional

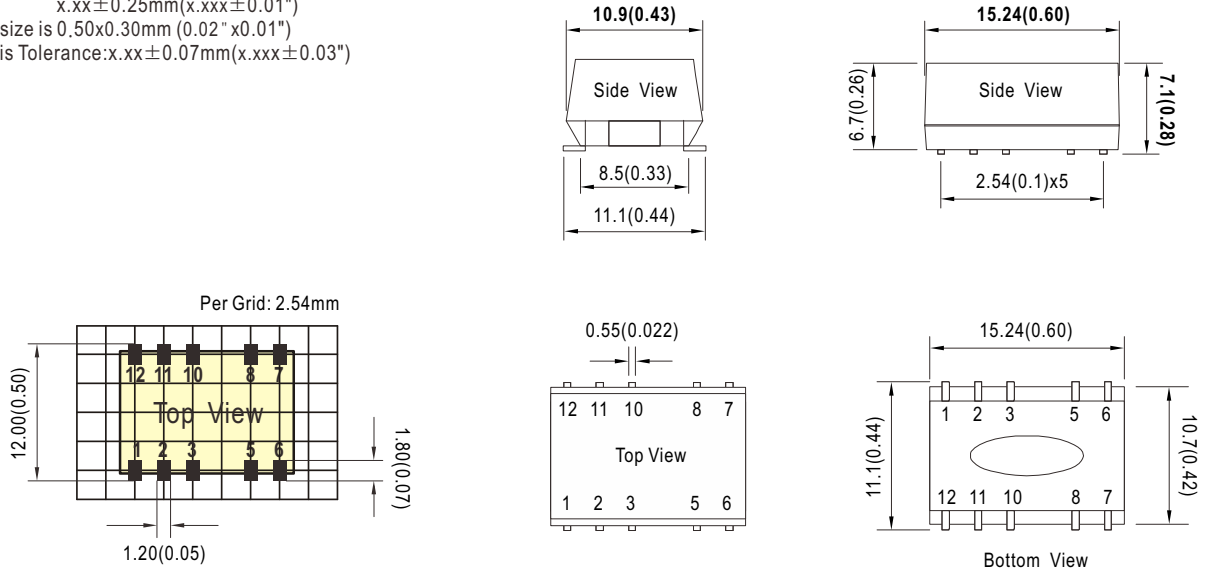


MODEL SELECTION TABLE							
ORDER NO.	INPUT			OUTPUT		EFFICIENCY (TYP.)	CAPACITOR LOAD (MAX.)
	INPUT VOLTAGE (RANGE)	INPUT CURRENT		OUTPUT VOLTAGE	OUTPUT CURRENT		
		NO LOAD	FULL LOAD				
DETN01L-05	Normal 5V (4.5 ~ 5.5V)	30mA	260mA	±5V	±10 ~ 100mA	78%	220μF
DETN01L-12		28mA	257mA	±12V	±4.2 ~ 42mA	78%	220μF
DETN01L-15		28mA	253mA	±15V	±3.4 ~ 34mA	80%	220μF
DETN01M-05	Normal 12V (10.8 ~ 13.2V)	13mA	107mA	±5V	±10 ~ 100mA	82%	220μF
DETN01M-12		12mA	103mA	±12V	±4.2 ~ 42mA	82%	220μF
DETN01M-15		13mA	102mA	±15V	±3.4 ~ 34mA	83%	220μF
DETN01N-05	Normal 24V (21.6 ~ 26.4V)	11mA	55mA	±5V	±10 ~ 100mA	76%	220μF
DETN01N-12		11mA	55mA	±12V	±4.2 ~ 42mA	77%	220μF
DETN01N-15		11mA	55mA	±15V	±3.4 ~ 34mA	78%	220μF

SPECIFICATION				
INPUT	VOLTAGE RANGE	L: 4.5 ~ 5.5Vdc M: 10.8 ~ 13.2Vdc N: 21.6 ~ 26.4Vdc		
	SURGE VOLTAGE (100ms max.)	5Vin models : 9Vdc 12Vin models : 16Vdc 24Vin models : 30Vdc		
	FILTER	Internal capacitor		
	PROTECTION	Fuse recommended. 5Vin models: 750mA Slow-Blow Type 12Vin models: 300mA Slow-Blow Type 24Vin models: 150mA Slow-Blow Type		
	INTERNAL POWER DISSIPATION	500mW		
	OUTPUT	VOLTAGE ACCURACY	±2.0%	
RATED POWER		1W		
RIPPLE & NOISE Note.2		75mVp-p		
LINE REGULATION Note.3		1.2% for 1% input variation		
LOAD REGULATION Note.4		±8%		
SWITCHING FREQUENCY (Typ.)		100KHz		
PROTECTION	SHORT CIRCUIT	Standard model: 0.5 second max. Optional models (SC-suffix): Continuous		
ENVIRONMENT	COOLING	Free-air convection		
	WORKING TEMP.	Standard model: -40 ~ +90℃ (Refer to "Derating Curve") ; Optional models (SC-suffix): -40 ~ +105℃		
	CASE TEMPERATURE	+100℃ max.		
	WORKING HUMIDITY	20% ~ 90% RH non-condensing		
	STORAGE TEMP., HUMIDITY	-55 ~ +125℃, 10 ~ 95% RH non-condensing		
	TEMP. COEFFICIENT	0.03% / °C (0 ~ 85℃)		
	SOLDERING TEMPERATURE	1.5mm from case of 1 ~ 10sec./240℃ max.		
	VIBRATION	10 ~ 500Hz, 2G 10min./1cycle, period for 60min. each along X, Y, Z axes		
SAFETY & EMC (Note.5,6)	SAFETY STANDARDS	UL62368-1, EAC TP TC 020/2011 approved		
	WITHSTAND VOLTAGE	I/P-O/P:3KVDC		
	ISOLATION RESISTANCE	I/P-O/P:100M Ohms / 500VDC / 25℃ / 70% RH		
	ISOLATION CAPACITANCE (Typ.)	80pF		
	EMC EMISSION	Parameter	Standard	Test Level / Note(Note.6)
		Conducted	BS EN/EN55032(CISPR32)	N/A
		Radiated	BS EN/EN55032(CISPR32)	Class B
	EMC IMMUNITY	Parameter	Standard	Test Level / Note
		ESD	BS EN/EN61000-4-2	Level 3, ±8KV air
		Radiated Susceptibility	BS EN/EN61000-4-3	Level 2, 3V/m
		EFT/Bursts	BS EN/EN61000-4-4	Level 1, 0.5KV
		Surge	BS EN/EN61000-4-5	Level 2, 0.5KV Line-Line
Conducted		BS EN/EN61000-4-6	Level 2, 3V	
Magnetic Field		BS EN/EN61000-4-8	Level 1, 1A/m	
OTHERS	MTBF	880Khrs min. MIL-HDBK-217F(25℃)		
	DIMENSION (L*W*H)	15.24*10.9*7.1mm (0.6*0.43*0.28 inch)		
	CASE MATERIAL	Non-Conductive black plastic (UL 94V-0 rated)		
	PACKING	Standard : 1.2g ; 32pcs/per tube, 3840pcs/120 tube/per carton Optional : 1.2g ; 500pcs/per reel, 2500pcs/5 reel/per carton		
NOTE	1.All parameters are specified at normal input(L:5Vdc, M:12Vdc, N:24Vdc), rated load, 25℃ 70% RH ambient. 2.Ripple & noise are measured at 20MHz by using a 12" twisted pair terminated with a 0.1μf & 47μf capacitor. 3.Line regulation is measured from low line to high line at rated load. 4.Load regulation is measured from 10% to 100% rated load. 5.The final equipment must be re-confirm that it still meet EMC directives. For guidance on how to perform these EMC tests, please refer to “EMI testing of component power supplies.”(as available on http://www.meanwell.com) 6.An external input filter capacitor is required if the module has to meet BS EN/EN61000-4-4, BS EN/EN61000-4-5. The filter capacitor Power Mate suggest: 470μF/100V. ※ Product Liability Disclaimer : For detailed information, please refer to https://www.meanwell.com/serviceDisclaimer.aspx			

Mechanical Specification

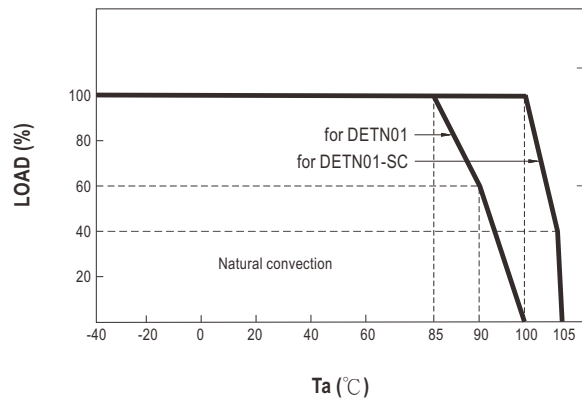
- All dimensions in mm(inch)
- Tolerance: $x.xx \pm 0.5\text{mm}$ ($x.xx \pm 0.02"$)
 $x.xx \pm 0.25\text{mm}$ ($x.xxx \pm 0.01"$)
- Pin size is $0.50 \times 0.30\text{mm}$ ($0.02" \times 0.01"$)
- Pin is Tolerance: $x.xx \pm 0.07\text{mm}$ ($x.xxx \pm 0.03"$)



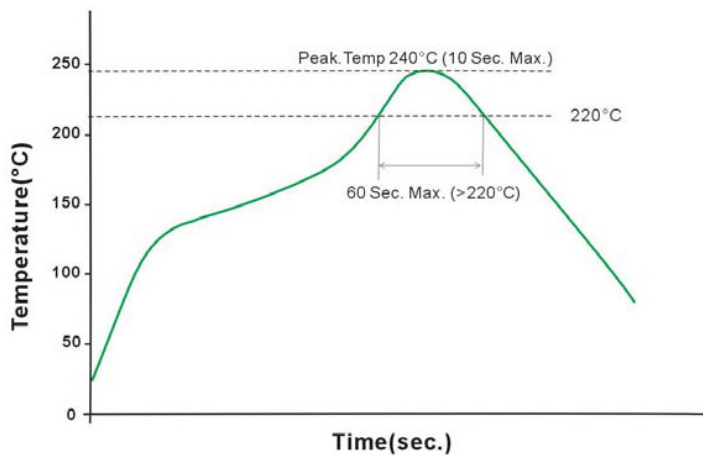
Plug Assignment

Pin No.	Pin-Out
1	-Vin
2	+Vin
5	Common
6	-Vout
8	+Vout
3,7,10,11,12	N.C.

Derating Curve

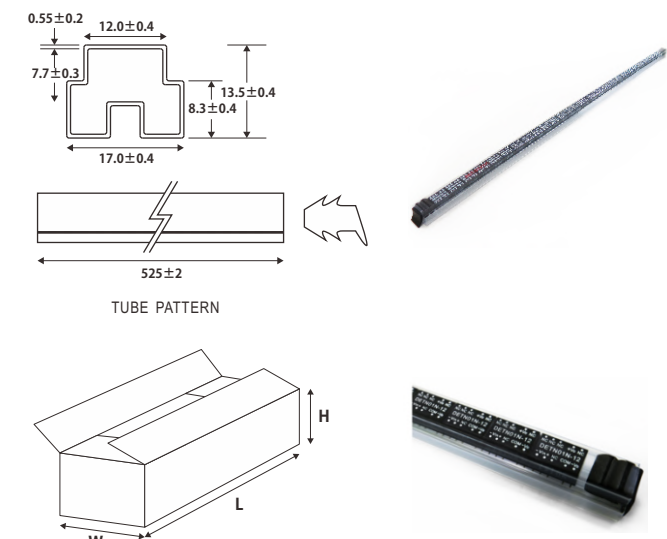
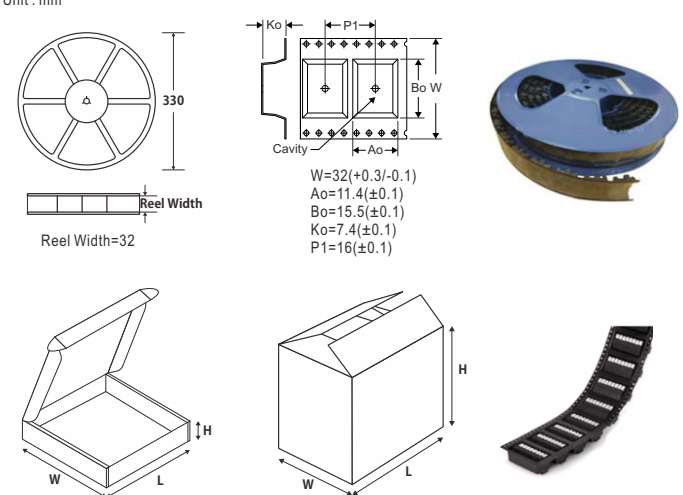


Reflow Soldering Curve



Remark: The curve applies only to the hot air reflow soldering.

Packing

Standard Tube Packing	MPQ Per Tube (PCS)	One Tube G.W.	Max. Q'TY/ Carton(PCS)	One Carton G.W.
Unit : mm  TUBE PATTERN CARTON L620 x W230 x H230	32	0.061Kg	3840	8.12Kg
Optional Reel Packing	MPQ Per Reel (PCS)	One Box G.W.	Max. Q'TY/ Carton(PCS)	One Carton G.W.
Unit : mm  Reel Width=32 INNER BOX L355*W338*H50 OUTER CARTON L380*W290*H370	500	1.3Kg	2500	7.3Kg

Installation Manual

Please refer to : <http://www.meanwell.com/manual.html>